

ECON 5345 Homework 8 Report

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(a) Combining the first three constraints, we have

$$Z_t K_{t-1}^\alpha L_t^{1-\alpha} = C_t + [K_t - (1 - \delta)K_{t-1}] + \frac{\phi}{2} \left(\frac{K_t}{K_{t-1}} - 1 \right)^2 K_{t-1}. \quad (1)$$

Then the maximization problem is

$$\max_{C_t, K_t, L_t} \mathbb{E} \sum_{t=0}^{+\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right\}, \quad (2)$$

subject to constraint (1).

The Lagrangian is

$$\begin{aligned} \mathcal{L} = \mathbb{E} \sum_{t=0}^{+\infty} \beta^t & \left\{ \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+1/\eta}}{1+1/\eta} \right] \right. \\ & \left. + \lambda_t \left[Z_t K_{t-1}^\alpha L_t^{1-\alpha} - C_t - [K_t - (1 - \delta)K_{t-1}] - \frac{\phi}{2} \left(\frac{K_t}{K_{t-1}} - 1 \right)^2 K_{t-1} \right] \right\}, \end{aligned} \quad (3)$$

where λ_t is the attached Lagrange multiplier. Taking FOCs with respect to C_t , K_t , and L_t , we get

$$[C_t]: \quad C_t^{-\sigma} = \lambda_t, \quad (4)$$

$$\begin{aligned} [K_t]: \quad & \lambda_t \left[\phi \left(\frac{K_t}{K_{t-1}} - 1 \right) - 1 \right] \\ & + \beta \mathbb{E}_t \lambda_{t+1} \left\{ \alpha \frac{Y_{t+1}}{K_t} + (1 - \delta) + \frac{\phi}{2} \left[1 - \left(\frac{K_{t+1}}{K_t} \right)^2 \right] \right\} = 0, \end{aligned} \quad (5)$$

$$[L_t]: \quad -\xi L_t^{1/\eta} + \lambda_t (1 - \alpha) \frac{Y_t}{L_t} = 0. \quad (6)$$

(b) Without stochasticity, we normalize $Z = 1$. Denote X^* as the steady state value of a variable X . We need to solve for $(C^*, K^*, L^*, I^*, Y^*)$, and the prices (R^*, W^*)

where R^* is the capital rental rate and W^* is the wage rate. Substituting (4) into (5) and (6), together with the constraints and the expressions for capital rental rate and wage rate, we get the system:

$$R^* + (1 - \delta) = \frac{1}{\beta}, \quad (7)$$

$$\xi(L^*)^{1/\eta}(C^*)^\sigma = W^*, \quad (8)$$

$$Y^* = (K^*)^\alpha(L^*)^{1-\alpha}, \quad (9)$$

$$Y^* = C^* + I^*, \quad (10)$$

$$I^* = \delta K^*, \quad (11)$$

$$R^* = \alpha \frac{Y^*}{K^*}, \quad (12)$$

$$W^* = (1 - \alpha) \frac{Y^*}{L^*}, \quad (13)$$

where (10) is derived by plugging the LoM. With 7 variables and 7 equations, the system is just-solvable.

- (c) Denote $\hat{x}_t = \frac{X_t - X^*}{X^*}$. Combining (4) and (5), we can log linearize the derived FOC of K_t to get

$$\begin{aligned} \mathbb{E}_t \hat{c}_{t+1} + \beta \left[\hat{k}_t [(\alpha - 1)R^* + (1 - \delta)K^* + \phi] - \mathbb{E}_t \hat{k}_{t+1} + (1 - \alpha)R^* \mathbb{E}_t \hat{\ell}_{t+1} \right] \\ = \phi(\hat{k}_{t-1} - \hat{k}_t), \end{aligned} \quad (14)$$

where (7) is used to substitute $R^* + (1 - \delta)$.

Combining (4) and (6), we can log linearize the derived FOC of L_t to get

$$(1 + 1/\eta)\hat{\ell}_t = -\sigma\hat{c}_t + \hat{y}_t. \quad (15)$$

The production function can be log linearized as

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_t + (1 - \alpha)\hat{\ell}_t. \quad (16)$$

The consumption-investment equation can be log linearized as

$$\hat{y}_t = \hat{c}_t \frac{C^*}{Y^*} + \hat{i}_t \frac{I^*}{Y^*}. \quad (17)$$

The LoM can be log linearized as

$$\hat{k}_t = (1 - \delta)\hat{k}_{t-1} + \hat{i}_t. \quad (18)$$

The prices can be log linearized as

$$\hat{r}_t = \hat{y}_t - \hat{k}_t, \quad (19)$$

$$\hat{w}_t = \hat{y}_t - \hat{\ell}_t. \quad (20)$$

The productivity can be log linearized as

$$\hat{z}_t = \rho\hat{z}_{t-1} + \varepsilon_t \quad (21)$$

Equation (16) can be used to estimate α . Equation (18) can be used to estimate δ .

Equation (14) can be used to estimate ϕ given the estimated α and δ .

Equation (21) can be used to estimate ρ and σ_ε .

Equation (15) can be used to estimate η and σ given the estimated σ .